5.4 Solving Rational Equations

A Rational Equations To solve a rational equation: ▪ State restrictions ▪ Multiply by the LCD (Least Common Denominator) ▪ Solve the polynomial equation (algebraically or by using technology) ▪ Verify restrictions ▪ Verify your solutions by substitution	Ex 1. Solve the following rational equation: $\frac{x}{x-2}-\frac{2}{x+3}=\frac{10}{x^2+x-6}$; $\frac{x}{x-2}-\frac{2}{x+3}=\frac{10}{(x-2)(x+3)}$ $\boxed{x\neq 2,-3}$ LCD : $(x-2)(x+3)$ $x(x+3)-2(x-2)=10$ $x^2+3x-2x+4=10$ $x^2+x-6=0$ $(x+3)(x-2)=0$ $x=-3,2$ $\therefore$ no solutions
B Cross Multiplication A rational equation of the form: $\frac{P(x)}{Q(x)}=\frac{R(x)}{S(x)}$, where $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ are polynomial functions, may be solve by cross-multiplication: $\frac{P(x)}{Q(x)}=\frac{R(x)}{S(x)} \Leftrightarrow P(x)S(x)=Q(x)R(x)$ Note. Do not forget to state and verify restrictions.	Ex 2. Use cross-multiplication to solve: $\frac{x-1}{2x+3}=\frac{x+2}{3x-2}$ $\boxed{x\neq -\frac{3}{2},\frac{2}{3}}$ $(x-1)(3x-2)=(x+2)(2x+3)$ $3x^2-5x+2=2x^2+7x+6$ $x^2-12x-4=0$ $x=6\pm\sqrt{40}$ $\boxed{x=6\pm 2\sqrt{10}}$ $\therefore x=6+2\sqrt{10},\ 6-2\sqrt{10}$
C Shortcut A rational equation of the form $\frac{P(x)}{Q(x)}=0$ is equivalent (if restrictions are satisfied) to the equation: $P(x)=0$ Note. Do not forget to state and verify restrictions.	Ex 3. Solve for x. $\frac{x^2+x-2}{x^2-1}=0$ $\frac{x^2+x-2}{(x-1)(x+1)}=0$ $\frac{(x+2)(x-1)}{(x-1)(x+1)}=0$ $x\neq 1,-1$ $\therefore \boxed{x=-2}$

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D No solution

A rational equation of the form:

$$\frac{constant}{P(x)} = 0$$

does not have any solution.

Ex 4. Solve for x.

$x \neq \frac{1}{2}, 1$

$$\frac{2}{2x-1} + \frac{1}{1-x} = 0$$

$$\frac{2-2x+2x-1}{(2x-1)(1-x)} = 0$$

$$\frac{1}{(2x-1)(1-x)} = 0 \quad \therefore \text{ no sol}$$

Ex 5. Solve for x.

a) $1 + x + \frac{4}{x} = \frac{9}{x-1}$

$x \neq 0, 1$

LCD $x(x-1)$

$x(x-1) + x^2(x-1) + 4(x-1) = 9x$

$x^3 + x^2 - x - x^2 + 4x - 4 - 9x = 0$

$x^3 - 6x - 4 = 0$

$2^3 - 12 - 4$

$-8 + 12 - 4 = 0$

$x = -2$

$$\begin{array}{r|rrrr} -2 & 1 & 0 & -6 & -4 \\ & & -2 & 4 & 4 \\ \hline & 1 & -2 & -2 & 0 \end{array}$$

$(x+2)(x^2 - 2x - 2) = 0$

$x = -2$

$x = 1 \pm \sqrt{1+2}$

$\therefore x = -2, 1 \pm \sqrt{3}$

b) $\frac{2x^3 - 1}{4 - x^2} = \frac{x}{x+2}$

$\frac{2x^3 - 1}{(2-x)(2+x)} = \frac{x}{x+2}$

$x \neq -2, 2$

LCD $(2-x)(2+x)$

$2x^3 - 1 = x(2-x)$

$2x^3 + x^2 - 2x - 1 = 0$

$x^2(2x+1) - (2x+1) = 0$

$(x-1)(x+1)(2x+1) = 0$

$\therefore x = \pm 1, -\frac{1}{2}$

Ex 6. Solve for x.

$x \neq 1, -1, 5$

a) $\dfrac{\frac{x}{x-1} + \frac{x}{x+1}}{\frac{3}{x+1} - \frac{2}{x-1}} = \frac{3x+2}{x-5}$

$$\frac{x\left(\frac{1}{x-1} + \frac{1}{x+1}\right)}{\frac{3x-3-2x-2}{(x-1)(x+1)}} = \frac{3x+2}{x-5}$$

$$\frac{\frac{x \cdot 2x}{\cancel{(x-1)}(x+1)}}{\frac{x-5}{\cancel{(x-1)}(x+1)}} = \frac{3x+2}{x-5}$$

$2x^2 - 3x - 2 = 0$

$x = \frac{3 \pm \sqrt{9+16}}{4}$ $\quad \frac{3+5}{4} = 2$, $\frac{3-5}{4} = -\frac{1}{2}$

$\therefore \boxed{x = 2, -\frac{1}{2}}$

b) $\frac{|x-1|}{x+1} + \frac{x+2}{|x-2|} = 3$

$x \neq -1, 2$

Case 1.

$x < 1$

$\frac{-x+1}{x+1} + \frac{x+2}{-x+2} = 3$

$(-x+1)(-x+2) + (x+2)(x+1) = 3(x+1)(-x+2)$

$x^2 - 3x + 2 + x^2 + 3x + 2 = 3(-x^2 + x + 2)$

$5x^2 - 3x - 2 = 0$

$x = \frac{3 \pm \sqrt{9+40}}{10}$ $\quad \frac{3+7}{10} = 1$, $\frac{3-7}{10} = \frac{-4}{10} = \boxed{-\frac{2}{5}}$

Case 2

$1 \leq x \leq 2$

$\frac{x-1}{x+1} - \frac{x+2}{x-2} = 3$

$x^2 - 3x + 2 - x^2 - 3x - 2 = 3x^2 - 3x - 6$

$-3x^2 + 3x + 6 = 0$

$x^2 - x - 2 = 0$

$(x+2)(x-1) = 0$

$\therefore x = 1, \cancel{-2}$

Case 3 $\quad x > 2$

$\frac{x-1}{x+1} + \frac{x+2}{x-2} = 3$

$x^2 - 3x + 2 + x^2 + 3x + 2 = 3x^2 - 3x - 6$

$x^2 - 3x - 10 = 0$ $\quad x = \frac{3 \pm \sqrt{49}}{2}$

$x = 5, \cancel{-2}$ $\quad \therefore \boxed{x = 5}$

$\therefore$ Sol are: $-\frac{2}{5}, 1, 5$

Reading: Nelson Textbook, Pages 278-285
Homework: Nelson Textbook, Page 285: #1, 2a, 3d, 5c, 6c, 7f, 13, 15

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